Universiti Tun Hussein Onn Malaysia

UNIVERSITI TUN HUSSEIN ONN MALAYSIA

FINAL EXAMINATION
SEMESTER I
SESSION 2022/2023

COURSE NAME	:	OBJECT ORIENTED PROGRAMMING
COURSE CODE	:	BIC 20904
PROGRAMME CODE	:	BIS / BIP / BIW / BIM
EXAMINATION DATE	:	FEBRUARY 2023
DURATION	:	3 HOURS
INSTRUCTION	:	1. ANSWER ALL QUESTIONS
		2.THIS FINAL EXAMINATION IS CONDUCTED VIA **CLOSED BOOK.**
		3.STUDENTS ARE **PROHIBITED** TO CONSULT THEIR OWN MATERIAL OR ANY EXTERNAL RESOURCES DURING THE EXAMINATION CONDUCTED VIA CLOSED BOOK.

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THIS QUESTION PAPER CONSISTS OF **FOUR (4)** PAGES

Q1 Based on program codes in **Figure Q1,** answer question **Q1(a) – Q1(d).**

```
class Number{
      private int b, total;

      void Count(int c) {
            int a=c;
            while (a<=6)
            {
                  b = a*a;
                  System.out.println(b);
                  total += b;
                  ++a;
            }

            System.out.printf ("Total is %d\n", total);
      }

      public static void main (String [] args) {
            Number x = new Number ();
            x.Count(2);
      }
}
```

Figure Q1

(a) Determine the output of the program.

(6 marks)

(b) Draw a class diagram based on the program codes in **Figure Q1**.

(5 marks)

(c) Based on the program codes in **Figure Q1** change the `while` loop statements into `for` loop statements.

(6 marks)

(d) Implement a constructor that will assign b with value 7 and display a statement "Numbers:" on the output screen.

(4 marks)

Q2 Write a Java program to read 5 integers and store them into an array. Then, write a method to find the largest and a function to find the smallest of them. Both methods must have a return data type. Finally, display the largest and smallest integer.

(29 marks)

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Q3 Based on program codes in **Figure Q3,** answer question **Q3(a) – Q3(b).**

```
class Oval{
      private int radius1, radius2;
      private double pi;

      Oval(){
            pi = 3.142;
      }

      public void setRadius1(int r1) {
            radius1 = r1;
      }

      public void setRadius2(int r2) {
            radius2 = r2;
      }

      public double getArea() {
            return pi*radius2*radius1;
      }
}
```

Figure Q3(a)

(a) Draw the UML diagram based on **Figure Q3(a).**

(9 marks)

(b) Based on **Figure Q3(a)**, write a Java program to create another class named `MainProgram`. In class `MainProgram`, implement main method and instantiate object from class `Oval`. Invoke all the required methods to calculate the area of Oval depicted as in **Figure Q3(b)** where the value of A and B are 77 and 88 respectively. The program must display the area of Oval on the output screen.

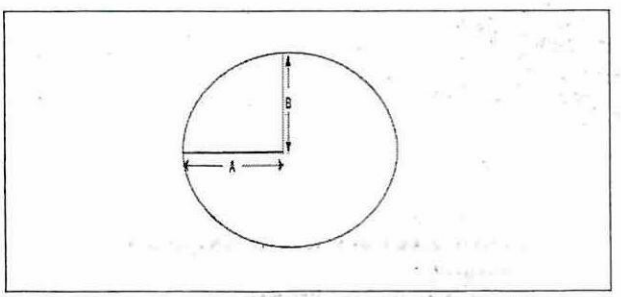


Figure Q3(b)

(6 marks)

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Q4 Based on program codes in **Figure Q4,** answer question **Q4(a) – Q4(e).**

```
public class XClass
{
      private int u;
      private double w;

      public XClass()
      {}

      public XClass(int a,double b)
      {}

      public void func()
      {}

      public String toString()
      {}
}
```

Figure Q4

(a) Write the definition of the method `func` so that `u` is set to `10` and `w` is set `15.3`.

(3marks)

(b) Write the definition of the method `toString` to return the values of `u` and `w` as a string.

(3marks)

(c) Write the definition of the constructor with the parameters of the class `XClass` so that the instance variable `u` is initialized to the value of `a` and the instance variable `w` is initialized to the value of `b`.

(3marks)

(d) Write a Java statement that prints the values of the instance variables of `objXClass`.

(3marks)

(e) Write a Java statement that creates the `XClass` object `temp` and initializes the instance variables of `temp` to `20` and `3.5`, respectively.

(3marks)

- END OF QUESTIONS –

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